

Aatman Mantri

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SUMMARY

Senior Software Engineer with 5+ years building and operating production backend systems in Python and Go. I design multi-service architectures, optimize high-traffic Django/DRF APIs (48 queries → 1), and manage Kubernetes infrastructure across GCP and Azure. Outside core work, I build agentic developer tooling on top of LLM APIs (Anthropic Claude, OpenAI) and the Model Context Protocol.

EDUCATION

New York University Tandon School of Engineering

Master of Science in Computer Science; CGPA: 3.6/4

New York, USA

June 2023

RV College of Engineering

Bachelor of Engineering in Computer Science & Engineering; CGPA: 8.1/10

Bengaluru, India

June 2019

EXPERIENCE

Senior Software Engineer

Nference, Inc.

Oct 2023 – Present

Bengaluru, India

- Designed and built a multi-service DICOM server from scratch in Go (API gateway, DICOMweb server, pathology service, LRU cleanup) with hot/cold storage orchestration, making 100M+ medical images accessible on-demand where only 1M were before. Deployed across 3 production environments (GCP + Azure).
- Rewrote the platform's most critical Django/DRF API endpoints — hit on every page load by 4,100+ users — reducing database queries from 48 to 11 (cold) / 1 (warm) via phased batching, BoolOr aggregation, polymorphic bypass, and a three-tier cache (request → Redis → DB).
- Simplified the AI Studio architecture from 5 data stores to 2 by removing ETCD, RabbitMQ, and BadgerDB (dead code and overengineering). Removed deprecated v1 APIs and 3 dead services from the codebase.
- Built end-to-end self-service signup flow with reCAPTCHA v3, admin approval, duration-based access expiry, and a transactional email system (10/10 mail-tester score). Shipped by business deadline for customer event.
- Turned an experimental DZI-to-DICOM conversion script into a production FastAPI service for pyramidal pathology image generation, integrated with the full imaging pipeline. Now used in customer sales pitches.
- Implemented zero-downtime blue-green deployments across Kubernetes environments, eliminating ~3 min of downtime per release. Diagnosed and fixed recurring pod OOM crashes traced to zombie slab memory from exec liveness probes.

Software Engineering Intern

Nference, Inc.

June 2022 – Dec 2022

Boston, USA

- Conceptualised and implemented a modified weighted PageRank algorithm to improve the ranking order of search results.
- Examined 3.9TB patient data for Amyloidosis pre-markers, halving doctors' preliminary analysis time through efficient insights.

Software Engineer

Nference, Inc.

July 2019 – July 2021

Bengaluru, India

- Built a distributed search engine serving queries over 30TB of medical data in 200ms with active-backup failover and automatic index updates.
- Engineered a distributed vector retrieval system using Intel MKL/CBLAS, achieving 100ms lookups across 3M 300-dimensional vectors.

TECHNICAL SKILLS

Languages: Python, Go, SQL, C

Frameworks: Django/DRF, FastAPI, Gin

Infrastructure: Kubernetes, Helm, Docker, GCP, Azure, GitHub Actions

Databases: PostgreSQL, MongoDB, Redis

Tools: Grafana, Kafka, gRPC, Git, Orthanc/DICOM, MLFlow

AI/LLM: Anthropic Claude, OpenAI, Cursor SDK, Model Context Protocol (MCP)

Side Projects: Cursor SDK agent skills (transcript mining, PR/blog generation), MCP server integrations (Atlassian, Sanity, Slack, Grafana), graphify per-repo code knowledge graph

Publication: Acute Coronary Syndrome Risk Prediction Model — [IEEE Xplore](#)